

Steps to Create and Use a Custom Image in Azure

1. Create a Resource Group

- Open the **Azure Portal**.
- Search for "**Resource groups**" in the top search bar and click **Create**.
- Enter the following:
 - **Name:** *(your preferred name)*
 - **Region:** *(any desired region)*
- Click **Review + create** and then **Create**.

2. Create a Virtual Network and Subnet

- Navigate to **Virtual Networks**.
- Click **Create**.
- Provide a name, select the resource group you created earlier, and configure an address space.
- Under the **Subnet** tab, create a new subnet.
- Review and create the VNet.

3. Create a Network Security Group (NSG)

- Go to **Network Security Groups**.
- Click **Create**.
- Assign it to your resource group and region.
- Define necessary inbound rules (e.g., allow SSH or HTTP).
- Review and create.

4. Create a Virtual Machine

- Go to **Virtual Machines** > **Create**.
- Select your resource group and region.
- Choose an OS (e.g., Ubuntu/Windows) and VM size.
- Make sure to:
 - Enable **public IP address** for remote access.
 - Attach the previously created **VNet**, **subnet**, and **NSG**.
- Complete admin credentials and other basic settings.
- Click **Review + create** > **Create**.

5. Connect to the VM and Install NGINX

- Connect to the VM using **SSH (Linux)**
- For **Linux**:

```
sudo apt update
```

```
sudo apt install nginx -y
```

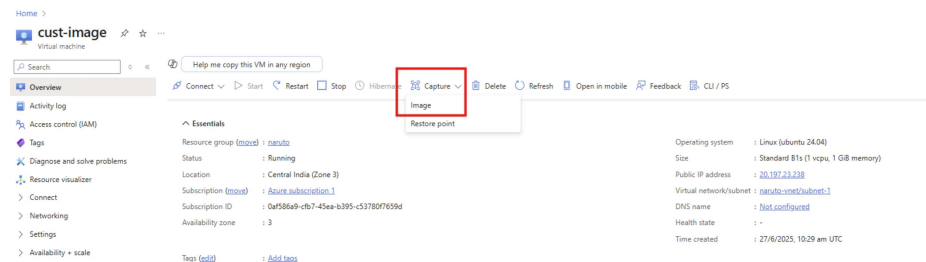
```
sudo systemctl enable nginx
```

```
sudo systemctl start nginx
```

- Ensure NGINX is running and accessible via the VM's public IP.

6. Capture the VM as a Custom Image

- Go to your VM in the Azure Portal.
- Click **Capture** from the top menu.
- If you plan to replicate this VM, consider locking the image.
- Fill in image details as shown in the pictures below and click **Create**.



Home > cut-image >

Create an image

Project details

Subscription

Azure subscription 1

Resource group *

nanuto

Instance details

Region

(Asia Pacific) Central India

Share image to Azure compute gallery

☒ Yes, share it to a gallery as a VM image version
 ☐ No, capture only a managed image.

Managed image is not available because it is not currently supported with Trusted launch virtual machines.

Automatically delete this virtual machine after creating the image

☐

Gallery details

Target Azure compute gallery *

(new) image

Create new

Operating system state

☐ Generalized: VMs created from this image require hostname, admin user, and other VM related setup to be completed on first boot
 ☒ Specialized: VMs created from this image are completely configured and do not require parameters such as hostname and admin user/password

Target VM image definition *

Create a VM image definition

Create new

Target VM image definition *

(new) nginx-image

Create new

Version details

Version number *

0.0.1

Exclude from latest

☐

End of life date

DD/MM/YYYY

Lock deleting Replicated Locations

☐

By unchecking this box, the ability to delete Image versions in replicated regions is unlocked.

Shallow replication

☐

Replication

A VM image version can be replicated to different regions depending on what makes sense for your organization. One example is to always replicate the latest image in multiple regions while all older versions are only available in 1 region. This can help save on storage costs for VM image versions.

Default storage sku

Standard HDD LRS

Default replica count *

1

Target regions	Replica count	Storage sku	Exclude from latest
(Asia Pacific) Centra...	1	Standard HDD LRS	☐
(US) East US	1	Standard HDD LRS	☐

Create a VM image definition

VM image definition name *

nginx-image

OS type

☒ Linux
 ☐ Windows

VM generation

☐ Gen 1
 ☒ Gen 2

Security type

Trusted launch

VM architecture

☒ x64
 ☐ Arm64

Higher storage performance with NVMe

☐

Hibernation supported

☐

Accelerated networking

☐

Publisher *

canonical

Offer *

ubuntu-24_04-lts

SKU *

server

Publishing options (Optional)

If needed replication then you need to lock it otherwise no need.

✔ Your deployment is complete

Deployment name

: Microsoft.Compute-CaptureVM-20250627161217

Subscription

: Azure subscription 1

Resource group

: nanuto

Start time

: 27/6/2025, 4:21:17 pm

Correlation ID

: 48767f29-0df6-418b-a33b-f76f6c2497b8

Deployment details

Resource	Type	Status	Operation details
✔ image/nginx-image/0.0.1	VM image version	OK	Operation details
✔ image/nginx-image	VM image definition	OK	Operation details
✔ image	Azure compute gallery	OK	Operation details

7. Create a New VM from the Custom Image

- Go to **Virtual Machines > Create**.
- Select **Image** > choose the custom image you created.
- Assign the same **VNet**, **subnet**, and **NSG**.
- Configure other settings as needed, then click **Create**.

Create a virtual machine

Help me create a low cost VM Help me create a VM optimized for high availability Help me choose the right VM size for my workload

Subscription * Azure subscription 1

Resource group * naruto

Create new

Instance details

Virtual machine name * image-vnet

Region * (Asia Pacific) Central India

Availability options Availability zone

Zone options Self-selected zone

Choose up to 3 availability zones, one VM per zone

Azure-selected zone (Preview)

Let Azure assign the best zone for your needs

Using an Azure-selected zone is not supported in region 'Central India'.

Availability zone * Zone 3

You can now select multiple zones. Selecting multiple zones will create one VM per zone. Learn more

Security type Trusted launch virtual machines

Configure security features

Image * image/nginx-image/latest - x64 Gen2

See all images | Configure VM generation

VM architecture x64

Arm64 is not supported with the selected image.

Microsoft Azure Upgrade Search resources, services, and docs (Ctrl+J) Copilot

Home > Compute infrastructure > Virtual machines > Create a virtual machine >

Select an image

Other Items | Shared Images

My Images

Shared Images

Community images

Direct Shared Images (PREVIEW)

Search in Shared Images

Publisher: All Azure Compute Gallery: All

Image Name	Subscription	Publisher	Gallery name
nginx-image	Azure subscription 1	canonical	image

8. Verify NGINX Installation

- Once the new VM is deployed, connect to it.
- Open the public IP in a browser or connect via SSH to confirm NGINX is installed and running.

image-vnet Virtual machine

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Connect

Networking

Settings

Availability + scale

Security

Essentials

Resource group (Group) naruto

Status Running

Location Central India (Zone 3)

Subscription (Group) Azure subscription 1

Subscription ID 0a586a9-cb7-454a-8385-c537807659a

Availability zone 3

Operating system Linux

Size Standard B1s (1 vCPU, 1 GiB memory)

Public IP address 20.197.23.238

Virtual network/subnet 10.0.0.0/24 / 10.0.0.0/24

DNS name Not configured

Health state ++

Time created 27/6/2023, 11:12 am UTC

```
root@image:~#
Broadcast message from root@cust-image (Fri 2025-06-27 10:50:34 UTC):
The system will power off now!
Connection to 20.197.23.238 closed by remote host.
Connection to 20.197.23.238 closed.
C:\Users\bsrut\Downloads>ssh -i keys.pem azureuser@20.197.0.244
The authenticity of host '20.197.0.244 (20.197.0.244)' can't be established.
ED25519 key fingerprint is SHA256:SA0y2w02LSjvYjHVOUCD79hIwAcE2DPMLdnuVQJ6w.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '20.197.0.244' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.11.0-1015-azure x86_64)

+ Documentation: https://help.ubuntu.com
+ Management:   https://landscape.canonical.com
+ Support:      https://ubuntu.com/pro

System information as of Fri Jun 27 11:14:47 UTC 2025

System load: 0.36          Processes: 119
Usage of /:  5.2% of 28.62GB Users logged in: 0
Memory usage: 31%         IPvd address for eth0: 10.0.0.5
Swap usage:  0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri Jun 27 10:36:04 2025 from 49.47.251.117
root@image:~# sudo systemctl status nginx
nginx.service - A high performance web server and a reverse proxy server
Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
Active: active (running) since Fri 2025-06-27 11:13:40 UTC; 2min 7s ago
Docs: man:nginx(8)
Main PID: 939 (nginx)
Tasks: 2 (limit: 995)
Memory: 3.6M (peak: 3.2M)
CPU: 23ms
CGroup: /system.slice/nginx.service
└─939 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
    └─942 "nginx: worker process"
```